

Syllabus

Villanova University

Course: CSC 4170 - Theory of Computation

Term: Fall 2025

Instructor: Dr. Maurício Gruppi

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Office: Mendel Hall, 292A

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Quick links

- [Course Schedule](#)
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Meeting times and locations

- **Section 001:** Tue/Thu 11:30 AM - 12:45 PM, Mendel Hall G88
 - **Section 002:** Tue/Thu 1:00 PM - 2:15 PM, Mendel Hall G88
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Office hours

- **Dr. Gruppi:**
 - Tuesdays 3 PM - 4 PM
 - Wednesdays 5 PM - 6 PM
 - *Note: You do not have to book a time to come to office hours, but I do appreciate if you let me know in advance that you will be coming.*
 - *Possibly available at other times or via Zoom upon request.*

**All office hours subject to change. Updates will be posted.*

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Main references

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- Textbook: "Introduction to the Theory of Computation" (3rd edition) By Michael Sipser. Cengage Learning, 2013. ISBN 978-1-133-18779-0
- Lecture notes and assignments will be available on the course LMS page.



Course content

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The following topics are covered in this course:

1. Regular languages: Finite automata, Nondeterminism, Regular Expressions, the pumping Lemma for regular languages.
2. Context-free languages: Context-free grammars, Chomsky normal form, pushdown automata, the pumping lemma for context-free languages.
3. Turing Machines: Definition, multitape TMs, nondeterministic TMs, enumerators, the Church-Turing thesis.
4. Undecidability: Undecidable languages, mapping reducibility.
5. Time complexity: The classes P and NP, NP-Completeness, P vs NP.
6. Advanced topic in complexity: approximation algorithms.



Course structure

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- Classes will take place in person at the designated time and location. The schedule will be available on Learning Management System (LMS) describing the contents of each class, including homework due dates, quizzes and exams.
- Evaluation will be based on quizzes and a final exam. Details in section **Assessment**.
- Office hours will be held to provide additional guidance to students. See section **Office hours** for details.

Course goals

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This course goes over what can and what cannot be achieved by computing. The goal is to expose students to the theoretical frameworks and concepts that lie at the core of computing.

Upon successful completion of the course, the student will be able to:

- Understand the aspects of regular languages, and how they are described by finite automata and its variants.
- Understand regular and context-free languages as well as its applications to computing.
- Understand the principles of Turing machines and how they act as the foundation to algorithms, computational complexity and computational thinking.
- Comprehend the theoretical aspects of computing, including the concepts of decidability and complexity theory.

Assessment

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Assessment of the learning goals in the class will be performed across various activities. Quizzes consist of 25% of the final grade, the final exam holds the remaining 75%. If advantageous to the student, these weights will be flipped after the final grade. That is, if you ace all the quizzes but do not perform very well on the final, your quizzes will be worth 75% and the final 25%, instead.

Grades will be distributed with the following activities:

Assignment	Points
Quizzes	25%
Final exam	75%
Homework	0%
Total	100

A description of each assessment module as well as their related goals is shown in Table 1. The final grade cutoff points are displayed in Table 2.

Assessment	Description	Goals assessed
Quizzes	Quizzes will be administered every Tuesday at the beginning of class. A quiz will be roughly composed of two problems based on the assigned set of homework questions.	Fundamental understanding, application, problem interpretation, problem solving, mathematical writing.
Final exam	A comprehensive exam with questions following a format similar to the quizzes.	Fundamental understanding, application, problem interpretation, problem solving, mathematical writing.
Homework	A problem set will be assigned as homework each week. Homework is not graded but it will work as a study guide in preparation for the weekly quizzes.	Fundamental understanding, application, problem interpretation, problem solving, mathematical writing.

Table 1: Assessment modules

Grade	Cutoff	Grade	Cutoff
A	95	C	65
A-	90	C-	60
B+	85	D+	55
B	80	D	50
B-	75	D-	45
C+	70	F	< 45

Table 2: Letter grade cutoff points

Attendance

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Attendance is expected for a successful completion of this course. The instructor will register the attendance at the beginning of every class.

As per university guidelines, if the number of unexcused absences for first year students exceeds twice the number of weekly classes (four absences for a course that meets twice a week), then such student will receive a failing grade “Y”.

Personal Days

In addition to the attendance policy stated above, students are entitled to one excused absence for any reason that may contribute to their personal wellness. Students must advise the instructor by email before class of their intent to utilize a Personal Day as the

reason for their absence. A Personal Day will not be approved retroactively. Students may, but are not required, to provide additional information regarding their absence. Additionally, a Personal Day may not:

- a. be used immediately preceding or following a University holiday or break period;
- b. be used on days when exams, presentations or other major assignments are scheduled.

A Personal Day does not grant an automatic extension for items due. Students remain responsible for all assignments, exams, presentations, etc. due on that date. It is in the instructor's discretion to determine whether any extension is appropriate given individual circumstances.

Technology

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- Students are allowed to make use of technology to assist them in the process of learning. Laptops, tablets and similar devices are permitted for uses related to the class and **should not cause distractions to anyone present**.
- All homework assignments **must be typeset**, unless otherwise stated. Students should make use of a computer and a text editor to write reports and homework solutions and export them in PDF format.
- Use of **Artificial Intelligence**: all submissions made by a student should be a result of their own work or, in the case of group work, their peers. The use of AI or Large Language Models such as ChatGPT, Microsoft Co-pilot, Llama 2 and Google Bard should be made with caution to ensure it will be beneficial to the student's learning outcomes. Answers generated by such models will **not be accepted** and will configure an **academic integrity violation**.

Late work

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Late work will not be accepted unless there is an excused absence. Make sure to notify the instructor as soon as possible to discuss the submission.

Student conduct

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As it is every member's responsibility to contribute to the creation of a healthy and safe community, students are required to comply with University health and safety directives, guidelines, rules, regulations and protocols in times of emergency and/or public health concern. Violations may be referred for action under the Code of Student Conduct.

Office of Disabilities and Learning Support Services

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It is the policy of Villanova to make reasonable academic accommodations for qualified individuals with disabilities. All students who need accommodations should go to Clockwork for Students via myNOVA to complete the Online Intake or to send accommodation letters to professors. Go to the LSS website or the ODS website for registration guidelines and instructions. If you have any questions please contact LSS at 610-519-5176 or learning.support.services@villanova.edu, or ODS at 610-519-3209 or ods@villanova.edu.

Academic Integrity

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All students are expected to uphold Villanova's Academic Integrity Policy and Code. Any incident of academic dishonesty will be reported to the Dean of the College of Liberal Arts and Sciences for disciplinary action. You may view the University's Academic Integrity Policy and Code for a detailed description.

Absence for Religious Holidays

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Villanova University makes every reasonable effort to allow members of the community to observe their religious holidays, consistent with the University's obligations, responsibilities, and policies. Students who expect to miss a class or assignment due to the observance of a religious holiday should discuss the matter with their professors as soon as possible, normally at least two weeks in advance. Absence from classes or examinations for religious reasons does not relieve students from responsibility for any part of the course work required during the absence. For more information, see Religious Holidays.

Lecture recording

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This course, including your participation, will be recorded on video and may be made available to students in the course for viewing remotely. Course videos and materials belong to your instructor, the University, and/or other sources depending on the specific facts of each situation, and are protected by copyright. Do not download, copy, or share any course or student materials or videos without the explicit permission of the instructor. For questions about recording and use of videos in which you appear please contact your instructor.

Copyright notice

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